

Matthew Topham

AI Systems Architect & Full-Stack Software Engineer

Boulder, Colorado | (805) 729-3307 | thetopham@gmail.com

github.com/thetopham | linkedin.com/in/mrtopham | thetopham.github.io

PROFILE

I build interactive AI products, persistent world systems, multi-node local AI infrastructure, and multimodal pipelines that turn AI roleplay into reference-grounded animated scenes. Active Marinara Engine developer with 100+ merged pull requests across React/TypeScript, Node.js, capability APIs, transactional state, AI generation, responsive UI, and production validation.

EXPERIENCE

Pasta Devs - Marinara Engine

2026-Present

AI Systems Architect & Open-Source Developer

- Authored 100+ merged pull requests across React/TypeScript UI, Node.js services, shared contracts, package APIs, transactional persistence, prompts, documentation, and automated tests.
- Shipped an end-to-end Roleplay-to-animation pipeline: completed exchanges become storyboard episodes; Character Sheets and World Maps ground people and locations; generated T=0 keyframes feed image-aware motion planning; local MiniMax H3/ComfyUI renders produced a live-tested multi-shot clip with synchronized motion and audio.
- Architected hierarchical world-state infrastructure with nested locations, graph validation, bounded prompt context, AI drafting/expansion, lore grounding, snapshots/checkpoints, revision conflict handling, and responsive map authoring.
- Extended the downloadable capability runtime with versioned host contracts for model routing, resources, persistence, transactions, history/checkpoints, readiness, retries, and package-owned client contributions.

AI INFRASTRUCTURE

Multi-node Local AI Fleet

- Built a reusable control plane across Linux and Windows hosts using SSH/Tailscale, PowerShell, and Windows Task Scheduler bridges for persistent remote workload control.
- Orchestrates logical server, ComfyUI, and AI workload roles from a central controller while keeping service-specific launch, stop, and status operations repeatable.
- Switches shared Radeon AI PRO R9700 compute between Qwen3.8 27B language-model serving and MiniMax H3/ComfyUI media generation instead of leaving competing workloads resident.
- Verifies service readiness using process state plus API health checks, including model/service endpoints, and passed repeated full workload-switch diagnostics ending with the GPU returned to idle.

TECHNICAL SKILLS

TypeScript | JavaScript | Python | SQL | React | Node.js | React Query | Vite | Postgres | REST APIs | WebSockets | Docker | GitHub Actions | Playwright | Linux | PowerShell | SSH | Tailscale | ComfyUI

AI systems: LLM integrations, agent/runtime architecture, local inference orchestration, reference-grounded storyboards, image-aware motion planning, local image/video generation.

SELECTED SYSTEMS & PROJECTS

PRIMARY / OPEN SOURCE

Marinara Engine + Agents

Local-first AI chat, roleplay, and game infrastructure with downloadable agents, persistent worlds, shared resources, and multimodal generation.

- 100+ merged PRs across Engine and Agents.
- Hierarchical world state, route validation, snapshots, transactions, recovery, World Maps, and Storyboard integration.
- Capability API 1.3 for package-owned model, storage, resource, and client services.

[Engine repo](#) [Agents repo](#)

SHIPPED PIPELINE

Roleplay to Animation

A complete media pipeline where everyday roleplay becomes a structured episode grounded by persistent visual references and rendered locally.

- Character Sheets and World Maps provide consistent people and location references.
- Image-aware shot planning refines motion against the generated first frame.
- MiniMax H3 / ComfyUI receives the grounded keyframe plus motion plan.

[Implementation](#) [H3 prompts](#)

AI / VR LEARNING

School of the Ancients

Voice-driven Socratic learning where users converse with AI historical mentors across web and XR prototypes.

- React/TypeScript interfaces, Node.js and Supabase/Postgres services, real-time voice AI, and Unity XR prototypes.

[Project](#)

AUTOMATION

AI Radar

Python and GitHub Actions pipeline that ingests AI research/news feeds, filters and summarizes them with LLMs, and publishes a concise daily brief.

[Repository](#)

EDUCATION & CERTIFICATION

University of Colorado Boulder

B.A. Computer Science - Expected May 2027

Front Range Community College

A.S. Computer Science - 2025

NVIDIA-Certified Associate

AI Infrastructure & Operations - 2025

LINKS

[GitHub](#) - github.com/thetopham

[Marinara Engine](#) - github.com/Pasta-Devs/Marinara-Engine

[Marinara Agents](#) - github.com/Pasta-Devs/Marinara-Agents

[Marinara contributions](#)

[Portfolio](#) - thetopham.github.io

[LinkedIn](#) - linkedin.com/in/mrtopham